

Research and application of hydraulic fracturing fluid entry depth detection method based on water-hammer signal

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ABSTRACT

Horizontal well staged fracturing method is a pivotal technology in the exploitation of unconventional oil and gas reservoirs. During the pump shut-in process, the water-hammer wave generated, which propagates along the wellbore and reflects at the fluid entry point and bridge plug. Pressure oscillation signals are observable at the wellhead during pump shut-in phase, which encapsulate information about the reflection period of the water hammer wave. By analyzing the pressure oscillation signal at the wellhead, it is possible to ascertain the depth of fluid entry in the process of hydraulic fracturing. In this paper, the water-hammer wave propagation model of wellbore and hydraulic fractures system is established and validated by using the field data. The relationship between wave velocity and reflection time is studied, finding the water hammer signal is a periodic resonance in the frequency domain. By picking up resonance frequency, the wave reflection period can also be obtained. In this paper, a novel frequency difference method (FDM) is proposed to calculate the reflection time in the frequency domain. The depth of the fracturing fluid entry point is determined by combining the echo time and wave velocity calculations. In this paper, field data from two field fracturing wells is collected. 10 staged water-hammer of wellhead pressure data are analyzed. The fluid entry point calculation results are consistent with the field results, which correlation coefficient is 99.6%. The results indicate the effectiveness of the proposed method. This paper provides a strong technical reference for determine fracturing fluid entry depth and hydraulic fracturing monitoring technology.

1. Introduction

Oil and gas are playing an irreplaceable role in promoting the economy, ensuring national security, and improving people's living standards (Zou et al., 2024; Yang et al., 2023). However, with the gradual depletion of conventional oil and gas resources and the increasing environmental problems, the development of unconventional oil and gas reservoirs has become particularly important. The increase in unconventional oil and gas production is due to the application of large-scale horizontal well staged fracturing technology (Yang et al., 2023; Zou et al., 2016), which forms a complex fracture (Liu et al., 2023; Saberhosseini et al., 2019). In the staged fracturing of horizontal wells, hydraulic fracturing can provide channels for underground oil and gas flow.

With the increase of the buried depth of the reservoir, the plasticity of the reservoir is enhanced, and the soil stress and its difference increase. In order to enhance oil recovery of the reservoir, it is necessary to add dense fractures. The number of clusters in a single segment is gradually increased from 2 to 3 clusters to 12–18 clusters. With the increase in the number of clusters, the clusters do not fully open in the actual fracturing process. The productivity of each cluster varies greatly. Whether these tightly-spacing fractures can be effectively created under the condition of strong stress interference remains unclear (Zhang et al., 2021). At present, fracturing fracture detection technology is difficult to evaluate the effectiveness of fracturing parameters economically and quickly, which limits the closed-loop optimization of fracturing effect.

The depth of fluid entry point can evaluate fluid entry different in the multi-cluster hydraulic fracturing. In addition, the depth of the fluid

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entry point can also help engineers optimize fracturing design, which helps to achieve more efficient fracturing operations. In summary, detecting the position of fluid entry point is a pivotal technology in the exploitation of oil and gas development efficiency and economic benefits.

The detection methods for the depth of the fluid entry point in fracturing treatment mainly include micro-seismic monitoring technology (Hison et al., 2005; Wang et al., 2024), optical fiber sensing technology (Holley and Kalia, 2015; Kalia et al., 2014), tracer technology (Kosynkin and Alaskar, 2016), acoustic or ultrasonic monitoring technology (Lee et al., 1994; Zhou et al., 2023; Yang et al., 2023) and so on. However, the above methods have some limitations, such as downhole operations risking, high equipment costs and susceptibility to external disturbances. For example (Lee et al., 1994), it is recognized that the micro-seismic energy is relatively small in their research. If the fractured reservoir is deep, the propagation of micro-seismic wave is obstructed, making it difficult for detectors arranged on the ground to receive micro seismic event. Holley and Kalia(2015) investigated the increase in fracturing production and found that ensuring the integrity of the fiber optic sensing system is the technical basis for all successful monitoring. Once the fiber optic is damaged in the wellbore, subsequent measurements will be affected by unpredictable serious consequences and even fail completely. This method depends large surface area observation system. Du and Guan, 2005 used over sixty kinds of oilfield tracers, the statistical results showed that chemical tracers such as nitrate and potassium iodide, were the most commonly used in field. Tang (2004) used acoustic logging data to process the imaging of out of well qualities, provided imaging examples, and preliminarily proved the feasibility of this technology.

It is worth noting that there is often a water-hammer effect during the fracturing treatment. At the end of each stage of fracturing construction, a sudden change in wellhead displacement will generate a water-hammer pressure wave in the wellbore and propagates along the wellbore. When it reaches underground hydraulic fracturing, a portion of the energy of wave is reflected back to the wellhead. By installing a high-frequency pressure signal acquisition device at the well head, the water-hammer pressure wave signal is collected during hydraulic fracturing. The water-hammer signal generated by pump shut-in contains rich underground information. By identifying and analyzing the fluctuation characteristics of wave signals, it can be used to evaluate the fluid entry position and scale of underground fractures (Hu et al., 2022). The method of diagnosing fractures based on water-hammer signals effectively utilizes the its effect formed during fracturing pump shut-in. This fracture diagnosis technology based water hammer is easy to operate, cost-effective, and has strong timeliness (Bogdan et al., 2016; Rutzinger et al., 2017). Specifically, after the staged fracturing and pump shut-in in a horizontal well, surface pressure acquisition equipment can record pressure fluctuations caused by water-hammer effects. By analyzing the propagation characteristics, such as velocity, reflection, and attenuation, the position, even size. The morphology of fractures can be identified (Klem et al., 2017; Dung et al., 2021; Carey et al., 2016). Dung et al. (2021) used water-hammer pressure data collected of the wellhead to obtain the pressure wave period, and studied the correlation between water-hammer characteristics and fracture detection results by assuming reasonable boundary conditions. In addition, Shen et al., (2023) studied the correlation between water-hammer signals and fracture characteristics, and found that due to the uncertainty of water-hammer velocity, inaccurate understanding of the propagation speed of water-hammer wave in the wellbore may lead to misjudgment of hydraulic fractures. In the actual process, the wave velocity is greatly affected by the parameters of the well, and reasonable calibration of the wave velocity is necessary. However, the above research results assume that the wave velocity is constant, and the research on the water-hammer wave has not been conducted. This means that the above methods may be limited in practical fracture diagnosis applications.

At present, some researchers have used specific characteristics in

water-hammer wave signals to determine the position of entry points (Hu et al., 2022; Ghidaoui et al., 2005; Parkhonyuk et al., 2018), and proposed analytical and numerical analysis methods for solving water-hammer differential equations. Hu et al. (2022) found that during field fracturing operations, the collected water-hammer signals generated by pump shut-in contain a large amount of noise. It is necessary to filter and denoise the water-hammer pressure signal in order to better analyze the fracture characteristics. Cao et al. (2019) solved the water-hammer equation using the method of characteristics and studied the effects of pump shut-in time and fracturing displacement on the water-hammer wave velocity. Similarly, Carey et al. (2015, 2016) and Hwang et al. (2017) discussed pressure signals on the wellhead and fracture detection results of horizontal fracturing, equivalently processing fracture parameters with circuits element. By adjusting the model parameters to match historical data, appropriate parameter ranges were obtained, providing reference parameters for fracture detection. To further increase the accuracy of numerical simulation, Ma et al. (2019) did not adopt the idea of equivalent circuit parameters of hydraulic fracture, directly considered perforation and fracture internal friction as boundary conditions. It is worth noting that in the above research results, most conclusions have made boundary assumptions in numerical calculations. When establishing a horizontal wellbore water-hammer model, the boundary conditions at the wellbore fracture bottom bridge plug node have been simplified, assuming that the fracture is located at the bottom of the wellbore. However, this simplified assumption may lead to computational errors in the model. In addition, the above results mainly focus on the analysis of the propagation characteristics of water-hammer wave in the wellbore, and there is no further analysis of the detection of entry point position. It should be pointed out that in actual fracturing analysis, the detection of the entry point position is one of the important parameters of horizontal well fracture diagnosis.

The main motivation of this article is to solve the detection of staged fluid entry points in horizontal wells based on wellhead pressure signal. The main innovations of this article are listed as follows: (a) The horizontal well wellbore fluid entry point bridge plug fracturing pump shut-in water-hammer model is constructed. Based on the transient flow theory, the characteristic line method is utilized to solve the model. By verifying the field-collected wellhead pressure data, the accuracy of the established model is demonstrated. On this basis, the influence of different factors on water-hammer response was analyzed, which provides a basis for fluid entry point detection. (b) The time-frequency domain characteristics of water-hammer signals are analyzed. An algorithm for locating the fracturing fluid entry point position is proposed, which uses the periodic method to calculate the wave velocity and the frequency difference method to calculate the echo time. By comparing the field application with the positioning results of the service provider's fluid entry point, the accuracy of the positioning method proposed in this paper is verified. (c) The process flow of locating the fluid point at the fracturing field is established, which can provide more accurate information on the entry point position under complex geological conditions, and provide important reference value for the evaluation of field hydraulic fracturing effects.

2. Methodology

In this section, we derive the fundamental equation of water-hammer effect based on the theory of transient flow. A mathematical model of water-hammer effect in horizontal well is established, which is solved by the method of characteristics. This paper adopts the equivalent circuit method to accurately describe the hydraulic fracture boundary in the horizontal well. The proposed model is then verified by field data. Finally, the influence of various factors, including entry point position and wave velocity on the model, are explored.

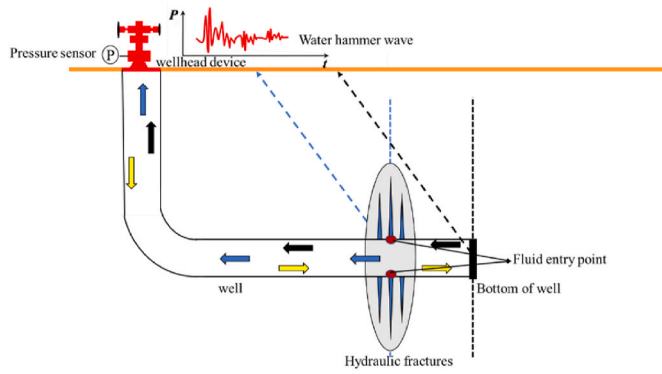


Fig. 1. Physical model of water-hammer signal detect fluid entry point position in horizontal well.

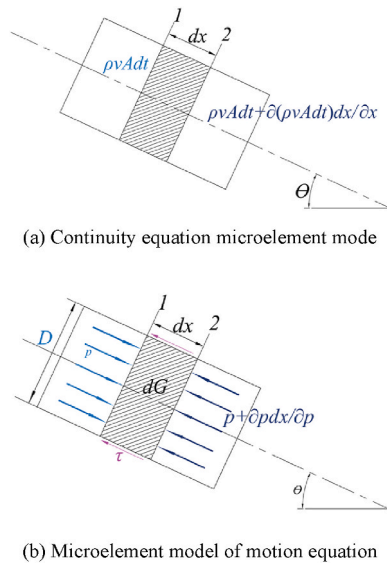


Fig. 2. Microelement model of continuity equation and motion equation.

2.1. Physical model of water-hammer

When the pump shut-in, the fluid velocity within the string will change rapidly due to the incompressibility and high inertia of the fluid. This rapid change produces sudden variations in stage treatment pressure, which is transmitted in the form of pressure wave. This characteristic is referred as water-hammer effect. The water-hammer occurs when the mechanical shock waves are generated at the fracture point in the bottom and the surface pressure sensor, leading to repetitive oscillations between the two points. The wellhead monitoring device captures the water-hammer pressure signal and analyzes it to determine the location of the fluid entry point. In this study, a physical model of water-hammer is presented, as illustrated in Fig. 1.

The fluid entry point refers to the precise position of the fracturing fluid underground. During multi-cluster fracturing operations, each cluster exhibits varying liquid inflows at this time, with the main fluid entry point identified as the cluster receiving the highest inflow. From this position, the critical information can be derived.

- (1) Through assessing the primary fluid entry point's depth, the length from the fracture to the wellhead can be calculated. This assessment allows for evaluating the efficacy of the fracturing treatment, thereby providing a field method for the fracture monitoring.

- (2) By locating of the primary fluid entry point, the effectiveness of the bridge plug can be further assessed, enabling the identification of bridge plug movement and whether it has been properly set.
- (3) Analyzing the fluid entry point enables optimizing the temporary plugging parameters, assessing the effectiveness of diverting methods, and achieving balanced initiation of multi-cluster fractures.

2.2. Water-hammer model assumptions

In each stage, the horizontal well multistage fracturing technology employs high-pressure fracturing apparatus to facilitate fracture propagation. The water-hammer effect initiated during the pump shut-in stage causes the pressure fluctuations within the well and fractures. The attenuation speed and duration of this effect depend on the fluid entry point. Current water-hammer models predominantly rely on the assumption of a constant pressure boundary, which fails to accurately depict the water-hammer characteristics within horizontal wells during the pump shut-in phase of fracturing treatment. This study aims to develop a mathematical model of water-hammer. Prior to presenting our primary findings, several assumptions are established. (a) The wellbore is modeled as a one-dimensional propagation; (b) The fluid within the wellbore is incompressible; (c) Physical properties, pressure and velocity are distributed at each point across the same cross-section; (d) The diameter of each part within the wellbore is constant, neglecting energy loss at the connection part; (e) The wellbore interacts with the formation exclusively through fractures; (f) The inclination of the horizontal section of the horizontal well remains constant.

2.3. Mathematical model of water-hammer

The fundamental principle of the water-hammer model based on the equations of mass conservation and momentum. Based on the theory of transient flow, the mathematical model of water-hammer is developed in this study.

Define the control volume 1–2 within the horizontal segment of the wellbore, as illustrated in Fig. 2(a). The fluid enters through section 1–1 and exits via section 2–2. The distance between these two sections is defined as, and the angle between the control volume's centerline and the horizontal plane is identified as. The cross-sectional area of section 1–1 is presented as A . Within the defined time interval, applying the mass conservation principle, the difference between the fluid mass entering section 1–1 and the fluid mass leaving section 2–2 is equal to the mass change within the control volume. This leads to the formation of the water-hammer continuity equation (Lipovsky and Dunham, 2015; Chaudhry, 2014)

$$\frac{\partial H}{\partial t} + v \sin \theta + \frac{a^2}{g} \frac{\partial v}{\partial x} = 0 \quad (1)$$

where v is fluid velocity, ρ denotes fluid density, H is the water head, a represents water-hammer wave velocity and g is gravity acceleration.

As illustrated in Fig. 2(b). Based on the principle of momentum conservation and assuming uniform flow velocity on the cross section of the pipeline, the motion equation for one-dimensional unstable flow can be derived using the average flow velocity on the cross section (Shen et al., 2023; Chaudhry, 2014).

$$\frac{\partial z}{\partial x} + \frac{1}{\rho g} \frac{\partial p}{\partial x} + \frac{1}{g} \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} \right) + \frac{4\tau_0}{\rho g D} = 0 \quad (2)$$

Where τ is the average stress on the cylindrical surface, z stands for the average elevation of the total flow section, $z = H - p/(\rho g)$, p represents the average pressure of the total flow section, D denotes the diameter of the total flow section. In equation (2), τ_0 is the shear stress of the pipe wall and $\tau_0 = \rho \lambda v |v|/8$, here, λ denotes the Darcy friction resistance. By

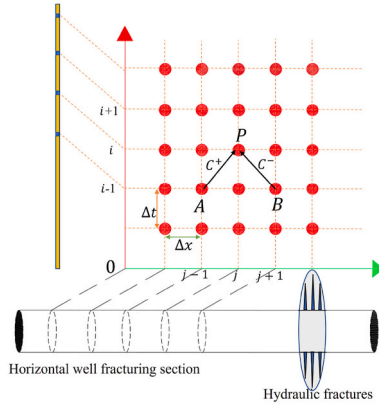


Fig. 3. Characteristic line grid diagram.

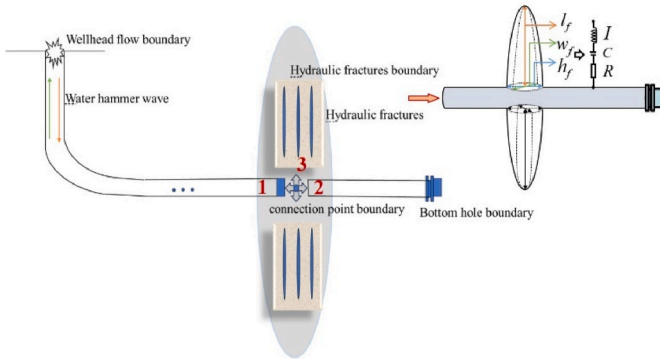


Fig. 4. Schematic diagram of boundary conditions.

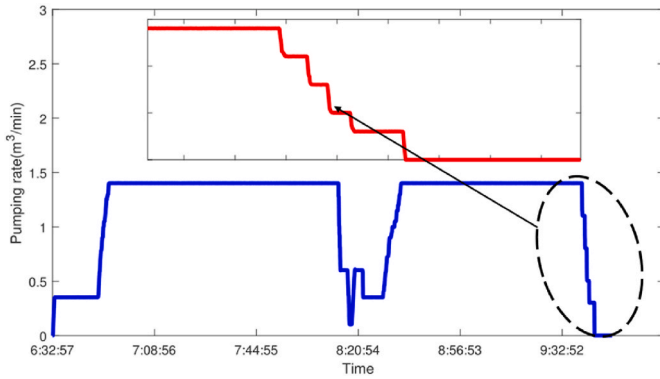


Fig. 5. Pumping rate of field data.

analyzing the range of Reynolds number Re (Chen, 1979), the hydraulic friction coefficient for the pipeline can be established via the appropriate empirical formula. The equation governing water-hammer motion considering friction resistance is obtained as follows (Chaudhry, 2014).

$$\frac{\partial H}{\partial x} + \frac{\partial v}{\partial t} + \frac{\lambda v |v|}{2D} = 0 \quad (3)$$

2.4. Method of characteristics

Using the characteristic line method to calculate the water hammer pressure at each point. As illustrated in Fig. 3, the water hammer pressure can be determined through these specific characteristic lines.

In this study, the first-order approximation method is adopted. The

Table 1
Basic stage treatment Information of well 1.

Parameter	Value	Parameter	Value
Well depth	5640 m	Depth of casing	6024.36 m
Depth of point A	3620 m	Horizontal segment length	2000 m
Casing diameter	139.7 mm	Casing thickness	12.7 mm
Young's modulus	39.851 GPa	Poisson's ratio	0.22
Ground elevation	377.51 m	Altitude	386.95 m
Minimum horizontal principal stress	85.1 MPa	Completion method	Casing completion
Initial resistance	89.87 bpd/psi	Initial fracture height	1.31064 m
Initial capacitance	2.44×10^{-3} bpd/psi	Initial fracture half length	122.9258 m
Initial inertance	3.09×10^{-7} psi/bbl/d2	Initial fracture Width	0.01524 m

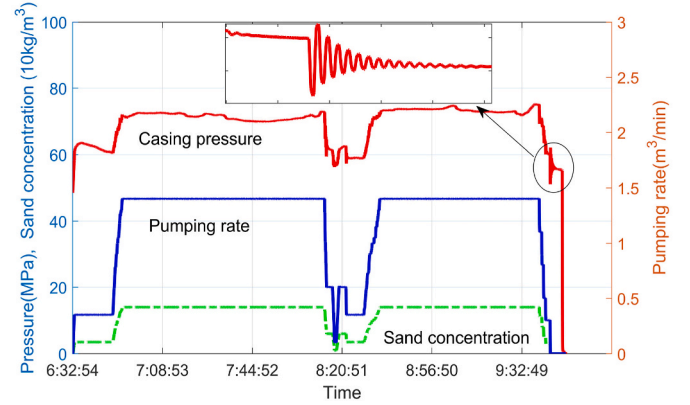


Fig. 6. Stage 20 fracturing operation curve of well 1.

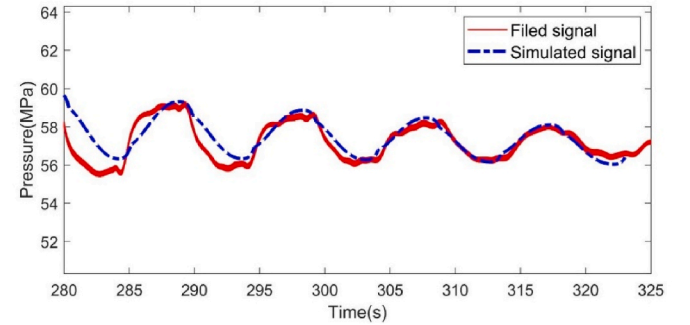


Fig. 7. Validation of proposed model.

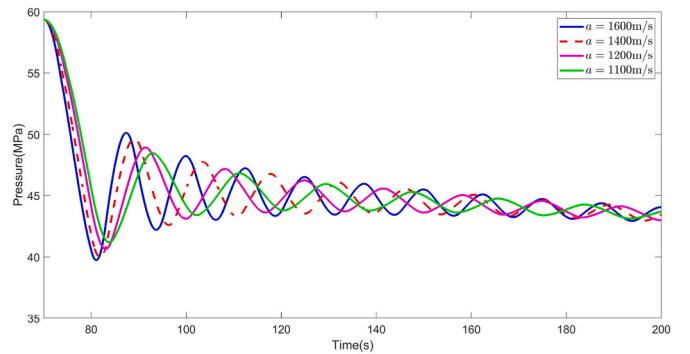


Fig. 8. Analysis chart of different wave velocity.

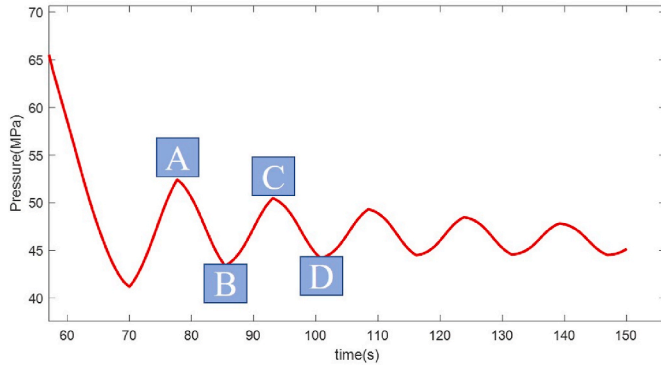


Fig. 9. Waveform of pump shut-in water-hammer with established model.

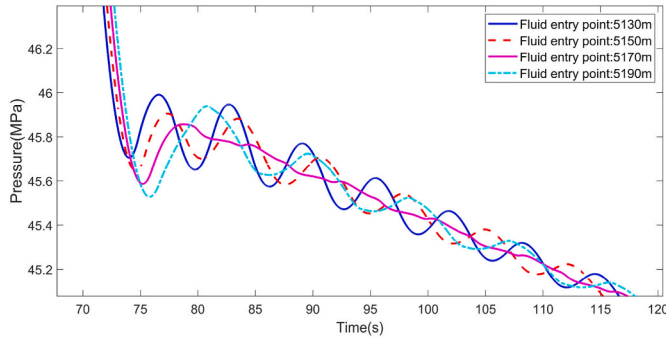
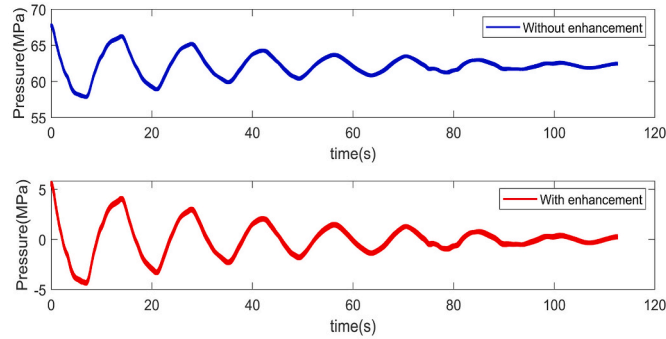
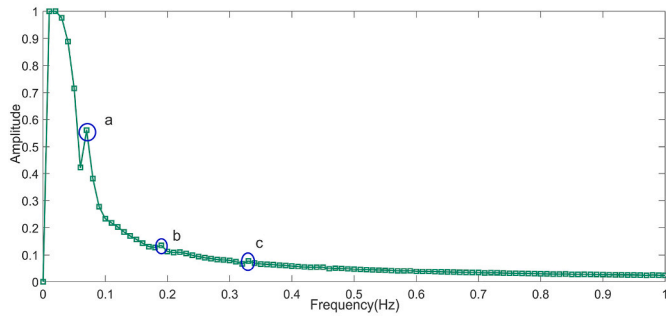


Fig. 10. The influence of fracture position on water-hammer wave.



(a) Time domain characteristic



(b) Frequency domain characteristic

Fig. 11. Time-frequency domain characteristic.

specific method is to use the known flow information of Q_A at point, or the flow Q_B at point B. Q_A or Q_B is used to replace the flow value in the integral term of these two equations, then the following results can be

Table 2

Calculation results of fluid entry point based on proposed method.

Type	Value	Unit
True value of wave velocity	1300	m/s
Calculated value of period	15.63	s
Calculated wave velocity	1296.3	m/s
Frequency difference	0.1295	Hz
True depth of fluid entry point	5000	m
Calculated depth of fluid entry point	5003.5	m

deduced (Chaudhry, 2014).

$$C^+ : H_p - H_A + B(Q_p - Q_A) + \frac{Q_A \Delta t}{A} \sin \theta + Z Q_A |Q_A| = 0 \quad (4)$$

$$C^- : H_p - H_B - B(Q_p - Q_B) + \frac{Q_B \Delta t}{A} \sin \theta - Z Q_B |Q_B| = 0 \quad (5)$$

where $B = a/gA$, $Z = (\lambda \Delta x) / (2gDA^2)$.

By replacing the nodes of time t and space i with the water head and flow identifications in the finite difference equation, the i point water head and flow at time $t + \Delta t$ can be calculated based on the $i - 1$ and $i - 1$ points conditions at time t . Thus, the above two equations can be organized as follows:

$$C_p = H_i^{t+\Delta t} + BQ_i^{t+\Delta t} = H_{i-1}^t + Q_{i-1}^t \left(B - \frac{\Delta t}{A} \sin \theta - Z |Q_{i-1}^t| \right) \quad (6)$$

$$C_m = H_i^{t+\Delta t} - BQ_i^{t+\Delta t} = H_{i+1}^t - Q_{i+1}^t \left(B + \frac{\Delta t}{A} \sin \theta - Z |Q_{i+1}^t| \right) \quad (7)$$

The characteristic line diagram indicates that once the horizontal well is discretized, only one single characteristic equation be applies at the grid's endpoint. Consequently, it is essential to establish an additional relationship between the head and the flow rate, i.e., the boundary conditions must be defined to meet the calculation requirements.

2.5. Boundary condition

The real model fractures and bottom bridge plugs of horizontal wells, as illustrated in Fig. 4 below. When constructing a water-hammer model for horizontal wells, be mindful of these following boundary conditions: (1) wellhead flow boundary; (2) connection point boundary between wellbore and fluid entry point; (3) fluid entry point boundary and, (4) bottom boundary.

2.5.1. Wellhead flow boundary

By equating the change in flow rate to a function of the change in the area of the valve opening over time, one can derive the connection between the flow rate at the wellhead (Carey et al., 2015).

$$\begin{cases} Q_1^t = Q_0 & t \in [0, t_1]; \\ Q_1^t = Q_0 \mu(t) & t \in [t_1, t_2]; \\ Q_1^t = 0 & t \in [t_2, +\infty); \end{cases} \quad (8)$$

where $\mu(t) = (1 - (t/T_1))^n$ is the opening coefficient, the value of which is between 0 – 1. $T_1 = t_2 - t_1$ is the valve closing time, n is the valve closing curve coefficient, if $n = 1$, it is a linear valve closing, Q_0 represents the initial flow.

During model validation, boundary condition 2 utilizes the field pump shut-in rate as the wellhead flow boundary condition. The corresponding curve is depicted in Fig. 5.

2.5.2. Wellbore and fluid entry point connection point boundary

The water head at the end of the horizontal section 1 is $H_{1,n}$, the water head at the beginning of the horizontal section 2 is $H_{2,1}$, and the water head at the beginning of the fracture is $H_{3,1}$. By disregarding head

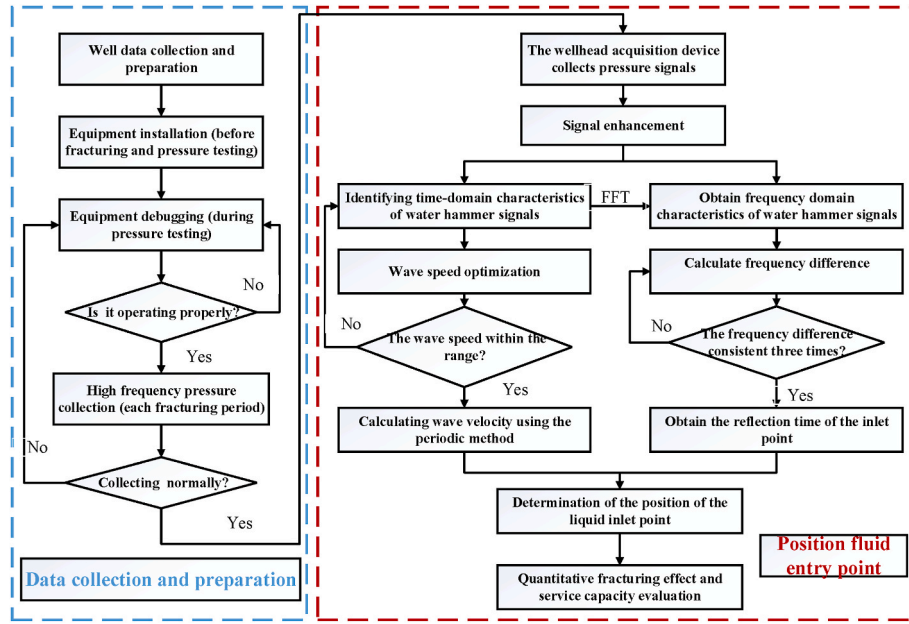


Fig. 12. Flow chart for the fluid entry point detection.

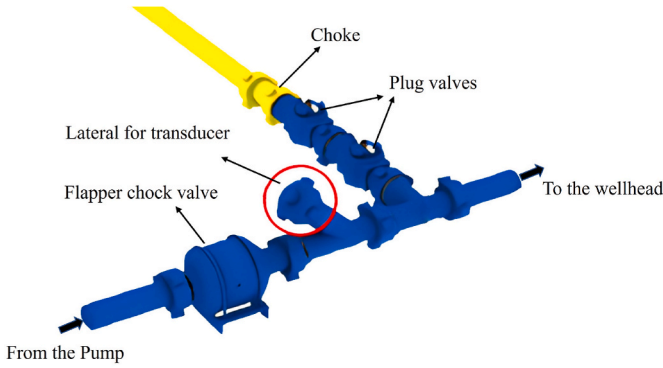


Fig. 13. Connection of special pressure sensor for high frequency pressure monitoring.

loss at the node, the water head of these three locations must be equivalent. Consequently, the flow relationship can be articulated as follows:

$$\begin{cases} H_{1,n} = H_{2,1} = H_{3,1} = H_t \\ Q_{1,n}^t - Q_{2,1}^t - Q_{3,1}^t = 0 \end{cases} \quad (9)$$

where

$$\begin{cases} Q_{1,n} = -\frac{H_t}{B_1} + \frac{C_{p1}}{B_1} \\ Q_{2,1} = \frac{H_t}{B_2} - \frac{C_{m2}}{B_2} \\ Q_{3,1} = \frac{H_t}{B_3} - \frac{C_{m3}}{B_3} \end{cases} \quad (10)$$

where the water flow at the beginning of the fracture is $Q_{3,1}$, the water flow at the beginning of the horizontal section 2 is $Q_{2,1}$, the water flow at the beginning of the horizontal section 2 is $H_{2,1}$, and the water head at the beginning of the fracture is $Q_{1,n}$.

The average water head can be determined by solving the two equations, which can be formulated as:

$$H_t = \frac{C_{p1}/B_1 + C_{p2}/B_2 + C_{p3}/B_3}{\sum (1/B)} \quad (11)$$

2.5.3. Fracture boundary

The conventional outlet conditions for water-hammer model typically establish a constant pressure boundary. However, the physical properties of the formation are not homogeneous, suggesting that this boundary does not reflect real-world scenarios. In this study, we employ the principle of equivalent circuit, as illustrated in Fig. 4. The fracture is modeled as an equivalent circuit composing resistance R , capacitance C and inductance I in series. Furthermore, the alteration in head and flow at the fracture junction can be represented by the circuit parameter R , C , I function.

$$\Delta H / Q = f(R, C, I) \quad (12)$$

This boundary condition outlines the effect of the fluid compressibility and inertia within the fracture. It also takes into account bending friction and perforation friction near the well area. When the pump stops, the pressure and flow rate changes of the fluid entering the fracture resemble the response in the series circuit. The fluid continues to flow only after overcoming the friction resistance in the fracture and reaching the minimum pressure necessary to maintain the fracture's openness. The correlation between pressure and flow can be articulated through a specific formula, demonstrated as equation (13).

$$P_f = \rho g \Delta H = RQ + \frac{1}{C} \int Q dt + I \frac{dQ}{dt} \quad (13)$$

Through the fracture friction equation, the bottom hole pressure is characterized as (Hu et al., 2022):

$$P_{BH} - P_f = P_{perf} + P_{nwf} \quad (14)$$

In this context, P_{BH} represents the bottom hole pressure, P_f indicates the average pressure within the fracture, P_{perf} refers to the friction resulting from perforation, and P_{nwf} denotes the friction caused by the bending of the wellbore in the proximity to the well area. The formula for calculating of perforation friction is provided below:

$$P_{perf} = \frac{8\rho Q^2}{\pi^2 N_p^2 d^4 C_D^2} \quad (15)$$

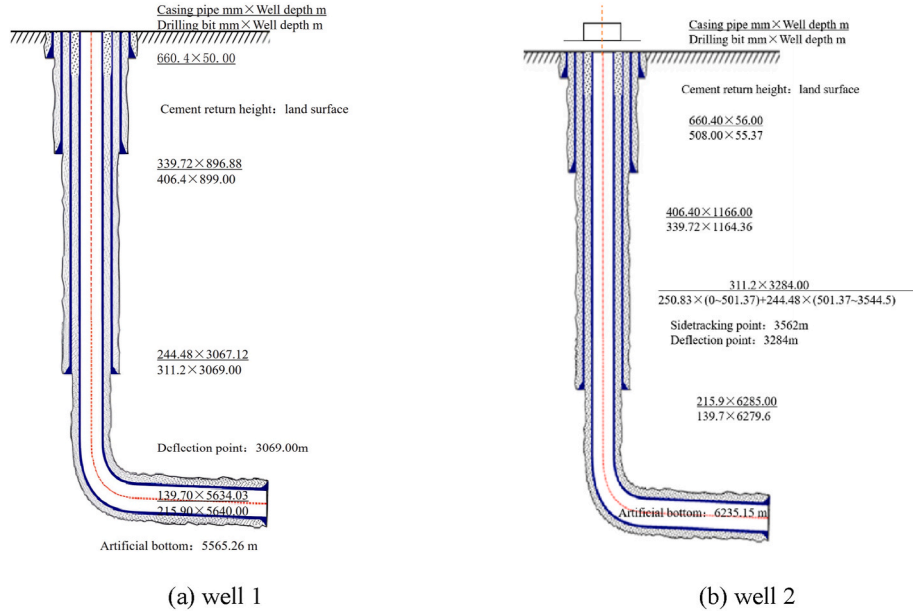


Fig. 14. Fracturing construction curve of well.

where N_p represents the number of perforations, and C_D is the flow coefficient, which is determined by the fluid type and the perforation size, and is regarded as a fixed constant. R denotes the friction coefficient near the well area. The friction force near the well area can be calculated using the following expression:

$$P_{nwf} = RQ^m \quad (16)$$

In general, the index m is established at 0.5 to represent the total impact of well bending on the flow resistance.

The inductance I accounts for the fluid's inertial effect, and its value correlates to the pressure difference required to either accelerate or decelerate the fluid $\Delta P = I \frac{dQ}{dt}$. Based on the comprehensive outline of the parameters and the corresponding circuit model, the pressure-flow relationship can be reformulated as follows (Song, 2024):

$$P_{BH} = \rho g H_n^{t+\Delta t} = R(Q_n^{t+\Delta t})^w + \frac{8\rho(Q_n^{t+\Delta t})^2}{\pi^2 N_p^2 d^4 C_D^2} + \frac{Q_n^{t+\Delta t} + Q_{n-1}^t}{2C} \Delta t + I \frac{Q_n^{t+\Delta t} - Q_{n-1}^t}{\Delta t} \quad (17)$$

For the fracture boundary conditions, the characteristic line equation can be used:

$$H_n^{t+\Delta t} + BQ_n^{t+\Delta t} = H_{n-1}^t + Q_{n-1}^t \left(B - \frac{\Delta t}{A} \sin \theta - Z |Q_{n-1}^t| \right) \quad (18)$$

2.5.4. Fracture dimensions from model parameters

After estimating the model parameters (R , C and I) through history matching of measured water-hammer data, these values can be used to calculate fracture dimensions. The equivalent circuit model parameter (R , C and I) are related to the fracture dimension (Sneddon, 1946; Shylapobersky et al., 1988):

$$w_f = \frac{\rho l_f}{I h_f} \quad (19)$$

$$l_f = \sqrt{\frac{4CI\Delta P}{\pi\rho}} \quad (20)$$

$$h_f = \sqrt{\frac{2G\alpha_s C}{l_f(1-\nu)}} \quad (21)$$

where, l_f , w_f , h_f respectively represent fracture height, fracture half length, fracture width. G is shear modulus, α_s stands for fracture shape factor, The value varies according to the fracture shape.

$$\alpha_s = \begin{cases} 0.63 + 0.425 \frac{h_f}{l_f}, & h_f > l_f \\ 0.63 + 0.425 \frac{h_f}{l_f}, & h_f < l_f \end{cases} \quad (22)$$

2.5.5. Bottom hole boundary

Because it is assumed that the wellbore only interacts with the surrounding strata through the fracture interface, the bottom hole is a closed end.

$$Q_n = 0 \quad (23)$$

2.6. Model validation

Based on the field water-hammer data, geological and engineering parameters, this study adjusts and optimizes the pertinent parameters in the existing simulation model. The goal is to ensure that the simulated water-hammer curve aligns closely with the field data. In this study, we utilize two sets of practical well field data, specifically the 20-stage fracturing stage treatment information from well 1 to validate our model. Details of are presented in Table 1, and the fracturing stage treatment curve is illustrated in Fig. 6.

This study begins with the input of initial conditions and basic parameters of geological engineering into the model. Subsequently, by taking several cycles of water-hammer oscillation windows for the water-hammer signal generated by fracturing pump shut-in, and continuously iterate to change the values of R , C , and I to match and compare the water-hammer waveform generated by fracturing pump shut-in with the actual pump shut-in water-hammer waveform until the numerical simulation results are similar to the filed pump-shut in water-hammer waveform, thereby the proposed model is validated, as shown in Fig. 7. The figure demonstrates that the proposed water-hammer model aligns well with the field water-hammer curve, exhibiting a maximum relative error of approximately 2.7%.

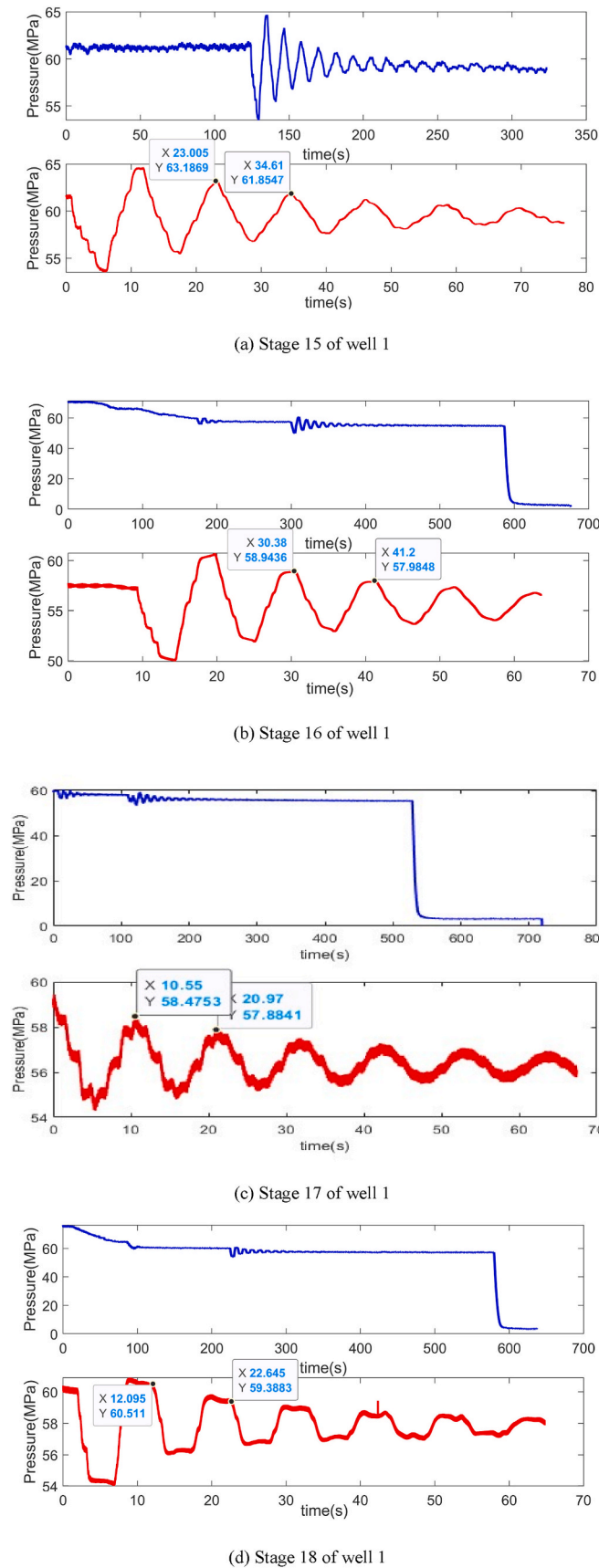


Fig. 15. Time domain extraction of periodic window water-hammer waveform in well 1.

Table 3

Different tested wave velocities of well 1.

Section number	Period (s)	Wave velocity (m/s)
15	11.4	1482.5
16	11.07	1493.4
17	10.445	1531.1
18	10.595	1492.1

3. Characteristic analysis of water-hammer effect

The water-hammer pressure wave produced by a fractured pump shut-in exhibits distinct characteristics in both the time domain and frequency domain. In the time domain, one can calculate the wave velocity based on the reflection period. Meanwhile, in the frequency domain, the location of the fluid entry point is determined by the frequency of the fundamental and the harmonic, i.e., the proposed frequency difference method in this study.

3.1. Time domain characteristic

In this subsection, the time domain analysis of water-hammer signals is mainly used to determine the wave velocity of pressure wave.

3.1.1. The influence of wave velocity

The influence of wave velocity on the internal water-hammer effect within the wellbore is vital, as it directly impacts both the magnitude and extend of the water-hammer effect, as well as its potential impact on the wellbore structure and fracture. Typically, wave velocity is governed by the fluid's compressibility and density inside the wellbore. In geological medium, the wave velocity is also influenced by the elastic modulus and density of the medium (Song et al., 2023). The precision of the wave velocity significantly affects the detection of the fluid entry point.

In this study, the wave velocity is assumed to be 1600 m/s, 1400 m/s, 1200 m/s and 1100 m/s, respectively. The impact of the wave velocity on the water-hammer signal is investigated, then different pressure-time curves are obtained, as shown in Fig. 8. The findings demonstrate that the larger the amplitude of water-hammer oscillation is, the more obvious the water-hammer effect is. The greater the amplitude of the water-hammer oscillation, the more pronounced the water-hammer effect, with a smaller oscillation period and slower attenuation. In contrast, at a water-hammer wave velocity of 1100 m/s, compared to the previous three wave velocities, there is a delayed reflection of the water-hammer signal, reduced oscillation amplitude, a notable oscillation period, and a quicker rate of attenuation.

The pump shut-in water-hammer signal curve recorded during the fracturing stage treatment is illustrated in Fig. 9. A and C in Fig. 9 constitute one cycle of water-hammer. The wellhead can be regarded as a closed boundary for the transmission of the water-hammer wave in the wellbore, while the bottom hole fracture boundary acts as an open boundary. The B and C moments denote the reflection of the expansion and the compression waves, respectively. The calculation for the reflection period T of the water-hammer pressure wave is outlined as follows $T = 4L/a$ (Hu et al., 2023), where L represents the distance from the wellhead to the bottom hole water-hammer reflection position, a signifies the water-hammer pressure wave velocity.

Based on the water-hammer model established above, this paper uses the periodic method to calculate the wave velocity, which is deduced as $a = 4L/T$.

3.1.2. The influence of the position of the fluid entry point

This study focuses on the actual staged and cluster perforation parameters of a well to examine how the distance between the entry point and the wellhead affects the water-hammer wave. The selected distance between the perforation site and the wellhead were set at 5130m,

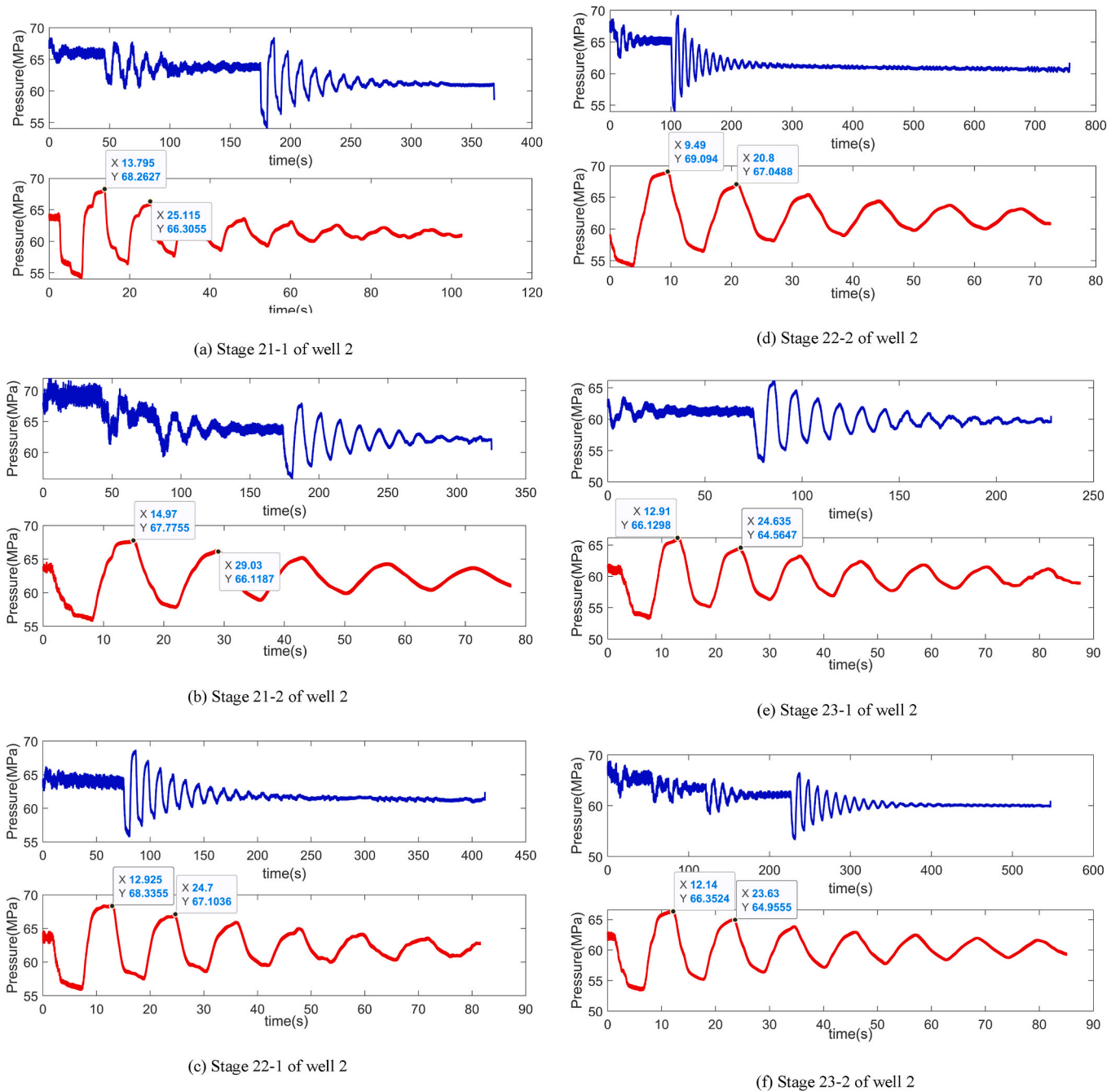


Fig. 16. Time domain extraction of periodic window water-hammer waveform in well 2.

Table 4

Different tested wave velocities of well 2.

Stage number	Period (s)	Wave velocity (m/s)
21-1	11.455	1500.5
21-2	13.83	1254.95
22-1	11.63	1443.85
22-2	11.51	1458.91
23-1	11.365	1447.6
23-2	11.145	1476.2

5150m, 5170m, 5190m, resulting in different pressure time curves, as shown in Fig. 10. The findings indicate that the water-hammer effect is more pronounced when the fluid entry point depth is situated near to the wellhead. Compared with the following three data, when the distance

between the fluid entry point and the wellhead is 5130m, the water-hammer wave is reflected back first, by analyzing the water-hammer signal at the wellhead, the position of the fluid entry point can be located.

3.2. Frequency domain characteristics

In this section, the analysis of water-hammer signals in the frequency domain is primarily utilized to ascertain the reflection time of pressure wave after reaching the fluid entry point. Initially, the time domain signal of water-hammer produced following the pump shut-in is converted from a pressure-time curve to amplitude-frequency curve in the frequency domain. Given the intricate high-frequency signal components of wellhead pressure, the noise component may compromise the precision of locating the fluid entry point, then research on pressure

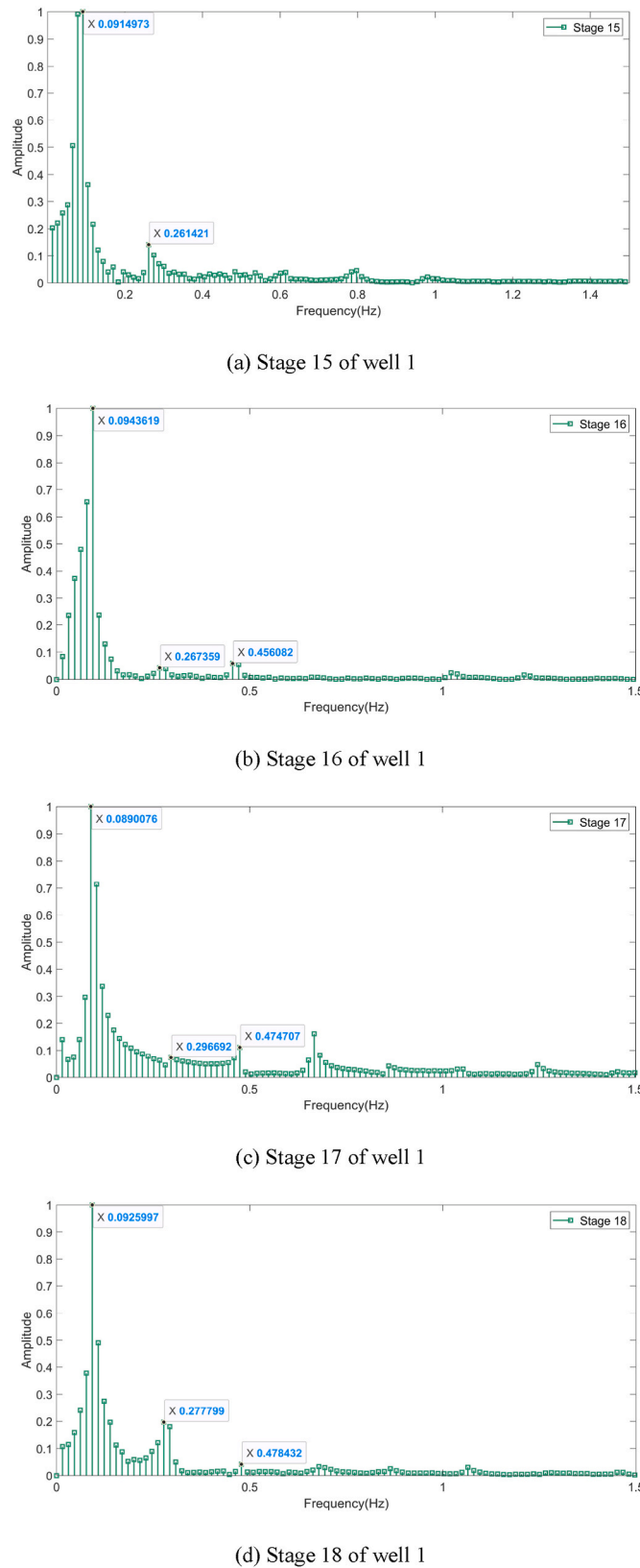


Fig. 17. Spectrum curve of well 1.

signal enhancement must be carried out.

In this study, the original time domain data of water-hammer pressure wave undergo trending to mitigate the amplitude of leakage frequency components, as depicted in Fig. 11(a). Following the detrending

Table 5

Different staged wave reflection time of well 1.

Stage number	Frequency difference (Hz)	Reflection time (s)
15	0.176	0.352
16	0.1849	0.3698
17	0.1916	0.3832
18	0.1881	0.3762

process, Fourier transform was performed to extract the frequency domain characteristics of the water-hammer signal, as shown in Fig. 11 (b). It can be seen from the frequency spectrum that there are multiple peaks, which are consistent with the reflection period of wave. Among them, the first peak is the fundamental frequency of the signal. In addition to the first fundamental frequency, the following peaks are the n th harmonic, and n is the serial number of the n th harmonic sequence. The boundary condition of the bottom hole fracture is the open boundary, and the fracturing fluid flows to the fracture. From the frequency spectrum, it can be concluded that the difference between the first-order harmonic frequency and the fundamental frequency is equal to the difference between the second-order harmonic frequency and the first-order harmonic frequency. Therefore, the reflection time of the water-hammer pressure wave generated by the pump shut-in can be calculated.

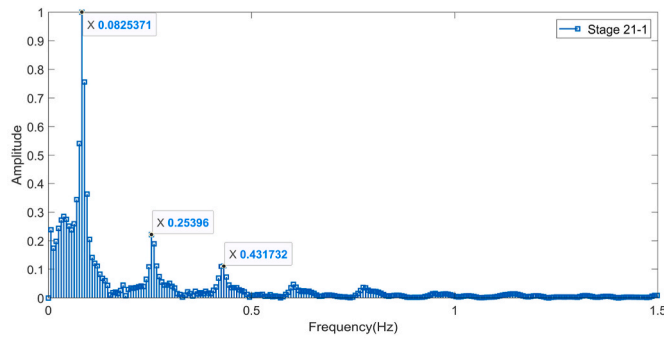
3.3. Fluid entry point calculation result with proposed model

Utilizing the time domain waveform derived from the water-hammer model, the wave period is calculated to determine the wave velocity, subsequently comparing with preset value. After signal enhancement, one can conduct a spectral analysis to ascertain the frequency differentials. Ultimately, one can compute the depth of the fluid entry point, and juxtaposed it with the theoretical position of the fluid entry point, with specific outcomes detailed in Table 2. The calculated wave velocity has an error of only 0.27% deviation from the true value, while the error of the position of the fluid entry point is 0.07% relative to the true value. The results indicates that the frequency difference methodology for determining the fluid entry point's position is not only viable but also highly accurate. Employing this positioning technique is advantageous for filed adjustment of fracturing treatment parameters during operational phase, thereby improving the effectiveness of transformation and establishing a robust groundwork for subsequent large-scale development of deep shale gas.

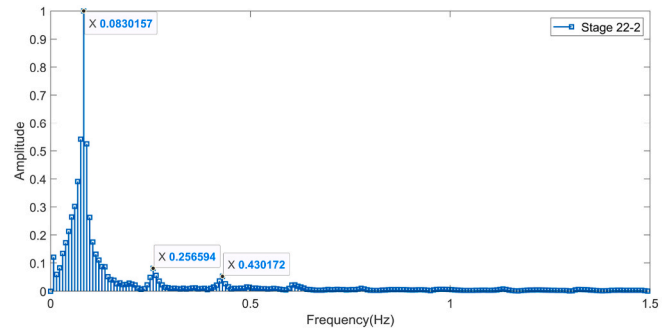
4. Processing method for detecting of fluid entry point

During pump shut-in, the pressure signal is captured using the high frequency pressure acquisition equipment for detecting fracturing fluid entry position. Utilizing the mechanical wave echo principle derived from the water-hammer effect occurring during pump shut-in, the wave velocity is determined by the periodic method, while the reflection time is assessed using a proposed frequency difference method. The position of the fluid entry point of the downhole fluid is analyzed in a short time, so as to achieve the purpose of quick decision in practical condition. The treatment process is shown in Fig. 12.

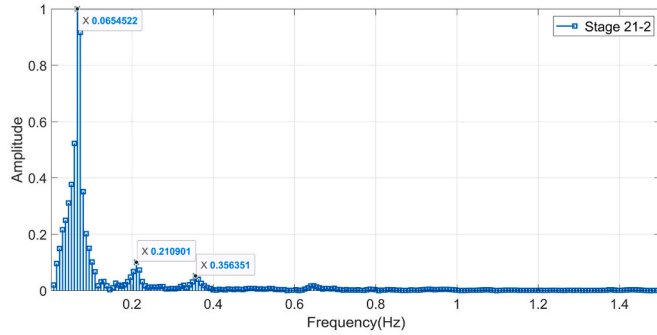
The high frequency pressure monitoring apparatus is illustrated in Fig. 13. From this illustration, it is evident that the arrangement of sensors, the connection and testing of acquisition lines, the monitoring of fracturing turning, and data processing are essential components. During the initial phase of equipment setup and commissioning for hydraulic fracturing operations, the high-pressure sensor interacts with the high-pressure pipe via a tee and a union fitting. The specialized pressure sensor is positioned between the wellhead valve and the single flow valve, ideally as close to the wellhead as possible. This connection method allows for continuous data collection after the pump is halted. Furthermore, the sensor is connected to the high-frequency pressure



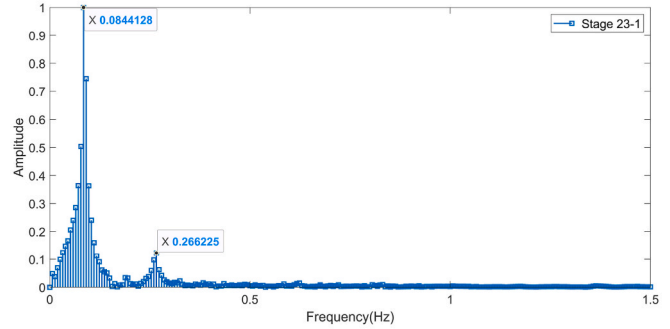
(a) Stage 21-1 of well 2



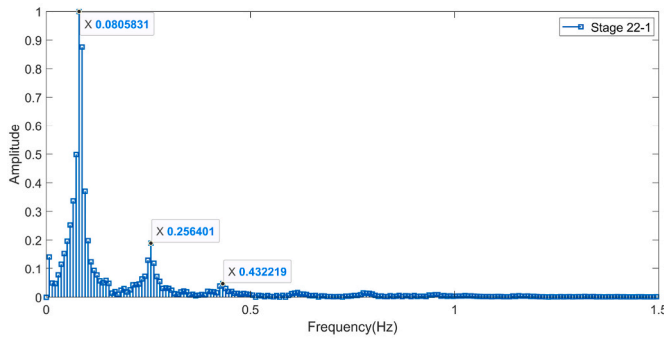
(d) Stage 22-2 of well 2



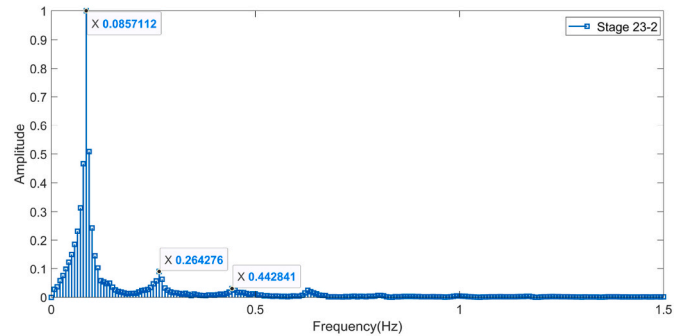
(b) Stage 21-2 of well 2



(e) Stage 23-1 of well 2



(c) Stage 22-1 of well 2



(f) Stage 23-2 of well 2

Fig. 18. Time domain extraction of periodic window water-hammer waveform in well 2.**Table 6**

Different staged wave reflection time of well 2.

Stage number	Frequency difference (Hz)	Reflection time (s)
21-1	0.1714	0.3428
21-2	0.1454	0.2908
22-1	0.1758	0.3516
22-2	0.1736	0.3472
23-1	0.1818	0.3636
23-2	0.1786	0.3572

collector via a data cable, and then connected to the interpretation computer through an output device for verification by comparing it with the pressure sensor data. Throughout the fracturing phase, the collection device continuously logs the pressure signal at the wellhead. Upon completion of each stage of stage, the collected data is remotely transmitted to the data interpretation center to ascertain the location of the fluid entry point and assist in assessing the hydraulic connection nodes

within the wellbore fracture system in real-world applications.

5. Field application

5.1. Basic information of well 1

The depth of Well 1 is 5640.00m and completed depth is 139.7 mm in diameter casing. The total length of the fractured stages in this well is 1963m, with each stage measuring between 90m and 98m, yielding an average length of 93.5m. There are 21 fractured stages in total, with cluster spacing ranging from 6.55m to 7.55m. The specific perforation parameters are as follows: stages 1–20 employ a single-stage 12-cluster perforation, with each cluster measuring 0.45m in length and comprising 7 holes; stage 21 utilizes a single-stage 9-cluster perforation, with the same cluster length and hole configuration. On the premise of ensuring the perforation number or perforation density, the perforation length can be modified. The wellbore configuration is illustrated in Fig. 14 (a).

Table 7

Results comparison between the service provider and the method proposed by this paper.

Stage number	Results provided by the service provider	Calculation results of the method proposed by this paper
Well 1-15	4160~4200m	4187.8m
Well 1-16	4060~4115m	4080.3m
Well 1-17	3975~4031m	3995.6m
Well 1-18	3873~3933m	3926.6m
Well 2-21-1	4275~4343m	4287.2m
Well 2-21-2	4260~4321m	4315.5m
Well 2-22-1	4176~4230m	4197.24m
Well 2-22-2	4190~4244m	4201.9m
Well 2-23-1	4089~4149m	4136m
Well 2-23-2	4082~4133m	4132.7m

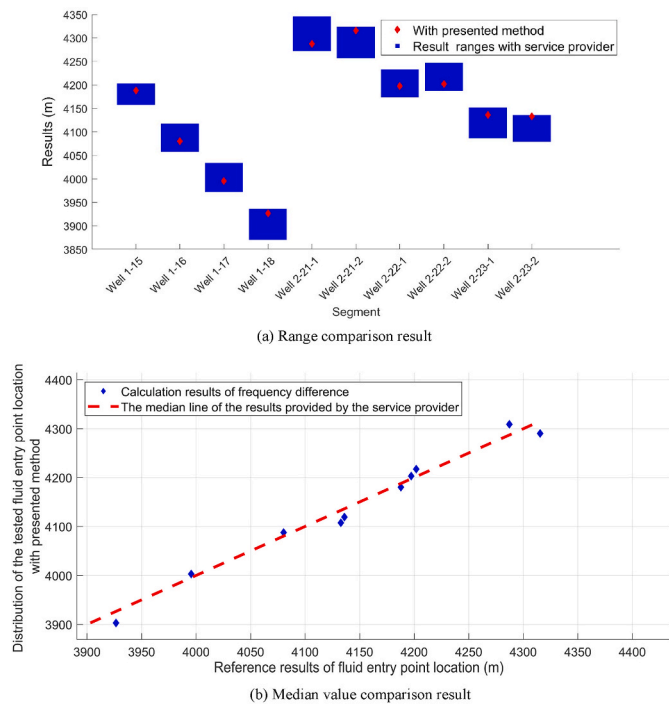


Fig. 19. Comparison results of the fluid entry point detection with the presented method.

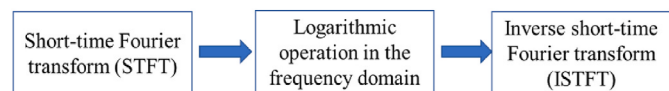


Fig. 20. Flow chart for the two-dimensional cepstrum calculation.

5.2. Basic information of well 2

The depth of Well 2 reaches a total 6285m. It has been completed with 139.7 mm casing. Given its high carbon content, it proves to be suitable for hydraulic fracturing applications. The length of the fracturing section measures 2159m, with each stage spanning from 79m to 101m, averaging 93.9m. There are a total of 23 fracturing stages, with the cluster spacing typically varying between 6.55m and 7.55m. The perforation parameters are as follows: stages 1–22 employ a single-

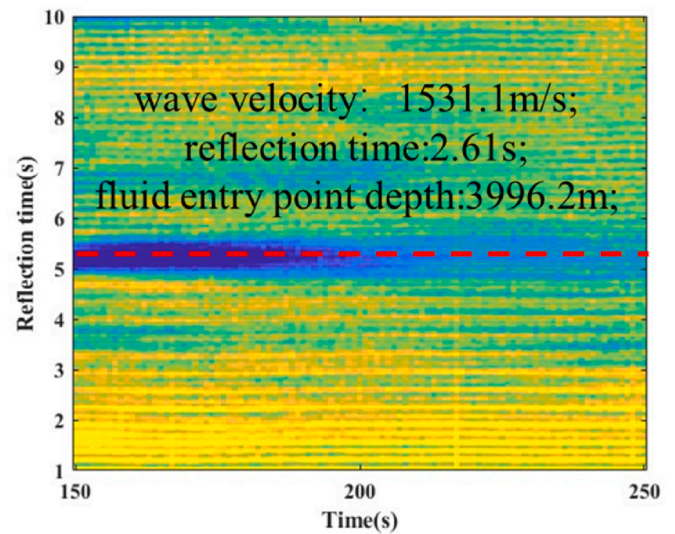


Fig. 21. Cepstrum results in stage 17 of well 1.

section 12-cluster perforation, where each cluster measures 0.45m in length and comprises 7 holes; stage 23 utilizes a single-section 11-cluster perforation, with each cluster being 0.45m in length and containing 7 holes. The wellbore structure is depicted in Fig. 14 (b).

5.3. Wave velocity calculation

In this subsection, the wave velocity of applied two wells are calculated with the presented method.

5.3.1. Wave velocity calculation for 15–18 stages of well 1

Here, we utilize the data of stages 15–18 to show our method. During the fracturing treatment of stage 15–18 of well 1, the wellhead pressure data is recorded at a frequency of 200Hz using a high-frequency pressure sensor and an advanced signal acquisition device positioned at the wellhead. Subsequently, the time domain waveform of the water-hammer pressure triggered by pump shut-in is windowed, as illustrated in Fig. 15. The reflection period is determined from the time domain representation of the water-hammer pressure, allowing us to compute the wave velocity for this stage.

By employing the periodic method to extract points from the time domain waveform of 15–18 stage, it is possible to derive the wave velocity across carious stages. The findings are presented in the subsequent Table 3.

5.3.2. Wave velocity calculation for stages 21–23 of well 2

Similarly, the signal at the wellhead with the time domain waveform of the water-hammer pressure produced by shutdown of the pump, being carefully sectioned. The findings are presented in Fig. 16, while the calculated wave data is summarized in Table 4.

5.4. Calculating reflection time added by frequency difference method

In this subsection, the reflection time of applied two wells are calculated with the presented method.

5.4.1. Calculating reflection time for well 1

The time domain waveform for stages 15–18 of well 1 was extracted by windowing in the previous section. A Fourier transform was utilized to the time domain pressure data to derive the spectral curve presented in Fig. 17. Employing the frequency difference method, the reflection times for stages 15–18 can be calculated, with the results displayed in Table 5.

5.4.2. Calculating reflection time for well 2

Through the time domain waveform before and after the temporary plugging of the stages 21–23 of well 2 obtained by the window in the previous section, the Fourier transform of the time domain pressure data is carried out, and the spectrum curve is shown as Fig. 18. Utilizing the proposed frequency difference method, the reflection times preceding and following the temporary plugging of stages 21–23 are calculated, as presented in the following Table 6.

5.5. Fluid entry point positioning analysis

In the above two subsections, the velocity of the water wave and the frequency in discrepancies are determined, respectively. The location of the entry point can be derived from Equation (24). To validate the precision and dependability of the calculation results, this paper compares the calculation results of the fluid entry point position based on the proposed method with the interpretations garnered from field monitoring. Additionally, the findings of well 2 further substantiate the alterations in the fluid entry point position before and after the temporary plugging. By assessing the effectiveness of temporary plugging, the fracturing parameters are optimized.

$$x = \frac{a}{2 \times \Delta f} \quad (24)$$

In this study, four sets of water hammer data related to pump shut-in at the conclusion of stage treatments from stages 15–18 of Well 1 were collected, and six sets of water hammer data from pump shut-in before and after temporary plugging during stages 21–23 of Well 2. The service provider uses an inversion-based two-dimensional cepstral method to locate the fluid entry depth (Parkhonyuk et al., 2018; Panjaitan et al., 2018). This study uses the frequency difference method to calculate the depth of the primary fluid entry point. The calculation results for both wells are summarized in Table 7. From the results, it can be seen that the frequency difference method yields the same results as those calculated by the service provider, indicating the correctness of the frequency difference method. The fluid entry points in stages 15–18 of Well 1 align with the actual detection results. Additionally, the calculation results indicate no signs of bridge plug detachment or intersection loss during the fracturing treatment, confirming that the intersection sealing is intact. The calculation results for the fluid entry points in stages 21–23 of Well 2 also correlate with the actual detection data, suggesting that the well interval is well sealed. New fluid is injected after temporary plugging fracturing, and the temporary plugging effect is clearly evident.

5.6. Results comparison with service provider

To ensure the accuracy and dependability of the fracturing fluid entry position in the field, the calculated entry position presented in this study is compared with the monitoring and interpretation results provided by the field service provider. The findings are illustrated in Fig. 19.

It can be observed from Fig. 19(a) that all the results derived in this study fall within the range of fluid entry points given by the service provider. It suggests that the calculation results of the method proposed in this paper are completely consistent with the monitoring interpretation results. Fig. 19(b) illustrates a comparative analysis between the median value of the fluid entry point range and the calculation results presented in this paper. From Fig. 19(b), it is evident that the average deviation between the two data sets is 19.57m, with a standard deviation of 16.82m, and the difference range spans from 5 to 25m. The high correlation coefficient of the two groups of data is 0.996, which signifies a robust linear relationship, indicating that their overall trends and variation patterns are very similar. The findings further substantiate the efficacy of the proposed method.

Currently, the frequency difference method for locating liquid entry

depth still has some limitations, which may be attributed to the following factors. First, the field water hammer signal is prone to noise interference, which affects wave velocity extraction and identification. Second, a low sampling frequency or inadequate signal processing resolution may result in the loss of critical details. Additionally, the precision and calibration of the sensors and equipment used for signal acquisition can affect signal quality and measurement accuracy.

To minimize these errors, the following measures can be implemented. Proposing more efficient water hammer signal noise removal techniques, which is also the direction the author intends to pursue in future research. Improving the existing wellhead pressure acquisition equipment to increase the sampling frequency of the water hammer signal at the wellhead. Selecting pressure sensors with faster response times, ensuring that the response time is kept within 1 ms.

5.7. Compared with cepstrum analysis

The two-dimensional cepstrum analysis requires segmenting the pressure signal by using a sliding window to compute the one-dimensional cepstrum within each window, and then stitching them together to obtain the two-dimensional cepstrum image (Hu et al., 2022). Furthermore, The two-dimensional cepstrum analysis can be broken down into three steps, as shown in Fig. 20.

This paper performs a cepstral analysis on water hammer data collected from stage 17 of well 1. Due to the negative impedance at the fracture, the fluid entry time corresponds to the larger negative impedance values in the time-domain cepstrum plot. According to Fig. 21, it can be observed that the dark blue band indicates the position where the negative cepstral value is the smallest, and the reflection time of the pressure wave at the fracture is 2.61s. In section 5.3, the wave velocity was calculated to be 1531.1 m/s, from which the fluid entry point depth was determined to be 3996.2 m. The error compared to the result calculated using the frequency difference method is 0.6 m, which proves that the method for calculating the fluid entry point depth proposed in this paper is accurate and simple.

The two-dimensional cepstral method analyzes water hammer signals to extract their periodic and frequency characteristics, providing high time-frequency resolution and strong adaptability. However, it requires extensive frequency domain transformations and complex signal processing during localization, which results in a high computational cost that may limit its real-time application. Additionally, the method demands high-quality raw signal data and is sensitive to parameters such as window size and overlap ratio. The selection of these parameters is often influenced by the experience of field engineer. Incorrect parameter settings of cepstral method can lead to significant errors, so precise tuning is essential to ensure accuracy.

In comparison, the frequency difference method determines the time and location of fluid arrival at the fracture by analyzing changes in frequency response. Compared to the two-dimensional cepstrum method, this approach is computationally simpler, requires fewer resources, is less influenced by variations in personnel experience, offers better real-time performance, and can quickly diagnose fracture locations, providing real-time guidance for optimizing field fracturing operations.

6. Conclusion

To address the problem of accurately detecting depth of fluid entry point, this paper proposed frequency difference method based on the high-frequency wellhead pressure signal.

Firstly, this study established a water hammer model of wellbore and fracture system. This model considered influence of wellbore that deeper than fluid entry point and bridge plug. This paper also adopted equivalent circuit model to describe resistance, capacitance and inductance characteristic of hydraulic fracture. The established water hammer model are validated by the field data. The error of model is

approximately 2.7%.

According to simulation results of above-mentioned model, the time and frequency domain characteristic of wellhead high-frequency pressure signal are analyzed. The wellhead pressure waveform is influenced by water hammer wave velocity and depth of reflection point. This paper proposed the periodic method to calculate the water hammer wave velocity. This wave velocity calculating method are verified by the numerical results. In the frequency domain, there are multiple peaks, which are consistent with the reflection time of fluid entry point. The depth of fluid entry point can be obtained by reflection time and water hammer wave velocity. This method can efficiently calculate the main fluid entry point based well head high-frequency pressure signal. The error of the positioning method of the fluid entry point is 0.07% relative to the true value by numerical simulation data. This study proposed signal processing of wellhead high-frequency pressure data. The process include Fourier transform and frequency difference picking. It can effectively ensure the accuracy of wellhead high-frequency pressure data interpretation.

The frequency difference picking method can efficiently obtain reflection time of water hammer wave. The calculation results are less influenced by human experience. According to downhole condition, the depth of main fluid entry point can be calculated by reflection time and wave velocity. This paper collected high-frequency pressure data of two field well. There are 10 stages water hammer pressure data of pump shut-in process. The sample frequency is 200Hz. The wave hammer wave velocity of 10 stages data are obtained by time domain oscillation wave. The reflection time of 10 stages data are obtained by frequency domain frequency difference. The depth of main fluid entry point of 10 stages are calculated by the obtained wave velocity and reflection time. The results of frequency difference method are compared with other service provider results. The correlation coefficient of the two results up to 0.996. This paper results are also compared with cepstrum analysis method. The error of two method is 0.6 m. However, the frequency difference method is more convenient to obtain depth of main fluid

entry points. The above research verifies the this paper proposed the calculation method.

The frequency difference method proposed herein demonstrates robust reliability. When compared with the results of the service provider and cepstrum analysis, the trend observed in this paper aligns closely with those results, indicating that the research outcomes hold significant value for optimizing operation plans and enhancing the hydraulic fracturing efficiency.

CRediT authorship contribution statement

Xuelian Dong: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Haiyan Zhu:** Investigation, Funding acquisition. **Xingming Wang:** Supervision, Funding acquisition. **Le He:** Software. **Ziping Liu:** Visualization. **Wei Gong:** Validation. **Li Zhan:** Visualization. **Jingran Tang:** Formal analysis. **Xiangyi Yi:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature list

dx	The distance between section 1-1 and section 2-2
A	The cross-sectional area of section 1-1
a	Wave velocity
C	Capacitance
C^+	Positive characteristic line
C^-	Negative characteristic line
C_D	The flow coefficient
D	The diameter of the total flow section
dA	The cross-sectional area of the micro-unit
dt	Time interval
E	Young's modulus
f	Friction factor
Δf	Frequency difference
g	Gravity acceleration
G	Shear modulus
H	Water head
$H_{1,n}$	The water head at the end of the horizontal section 1
$H_{2,1}$	The water head at the beginning of the horizontal section 2
$H_{3,1}$	The water head at the beginning of the fracture
H_t	The average water head
ΔH	The change in hydraulic potential
h_f	Fracture height
I	Inductance
i	The nodes of space
L	The distance from the wellhead to the bottom hole water-hammer reflection position
l_f	Fracture half length
m	Index

N_p	The number of perforations
n	The valve closing curve coefficient
p	Average pressure of the total flow section
P_f	The average pressure within the fracture
P_{BH}	The bottom hole pressure
P_{perf}	The friction resulting from perforation
P_{nwf}	The friction caused by the bending of the wellbore in the proximity to the well area
Q	Flow rate
Q_A	Flow rate at point A
Q_B	Flow rate at point B
Q_0	The initial flow
$Q_{3,1}$	The flow rate at the beginning of the fracture
$Q_{2,1}$	The flow rate at the beginning of the horizontal section 2
$Q_{1,n}$	The flow rate at the end of the horizontal section 1
R	Resistance
t	The nodes of time
T	The reflection period
T_1	The valve closing time
w_f	Fracture width
x	The fluid flow direction
z	The average elevation of the total flow section
α_s	Fracture shape factor
θ	The angle between the control volume's centerline and the horizontal plane
λ	The Darcy friction resistance
v	Fluid velocity
ρ	Fluid density
τ	the average stress on the cylindrical surface
τ_0	the shear stress of the pipe wall
ω	eigenvalue

Data availability

No data was used for the research described in the article.

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